

Advanced Microeconomics

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Problem Set 10: Recap Theory and Applications of General Equilibrium

1 Recap Marginal Value of Public Funds and Value of Statistical Life

1) Answer the following questions:

- a) Briefly describe the concepts of willingness-to-pay (WTP) and Marginal Value of Public Funds (MVPF).

Solution: The WTP is the maximum amount of money an individual is willing to spend for a product or service. This reflects how much the product or service is worth to the individual. Ways to calculate this amount of money are surveys, experiments or auctions. In the context of the MVPF the WTP represents the amount of money individuals are willing to pay for the policy change, and thus how much the policy change is worth to them. For one dollar cash transfer a rational individual should be willing to pay (at maximum) one dollar.

The MVPF is constructed as follows:

$$MVPF = \frac{WTP}{Net\ Government\ Costs}$$

The MVPF thus is the ratio of how much the policy is worth to the individuals who profit from it (WTP) to the policy's costs that the government has to bear. In addition to the WTP the government costs are needed to calculate the MVPF. The government has to bear two types of costs: the actual costs of the policy change (Mechanical Costs) and the costs which arise due to changes in behavior of the individuals due to the policy change (Fiscal Externality). The MVPF is usually normalized to one dollar in mechanical costs. Therefore a MVPF of \$1.5 means that for each dollar of mechanical costs for the government beneficiaries are willing to pay up to \$1.5.

- b) A study reports MVPFs of changes of the Earned Income Tax Credit with values at 1.2 and 1.12, while other cash transfers have MVPFs below 1. Interpret these differences and give an intuition for why they occur.

Solution: If the MVPF is normalized to one dollar of mechanical costs the values can be interpreted as follows. If $MVPF > 1$, this means that a \$1 increase in mechanical expenditure on the policy either increases the benefits of the beneficiaries by more than 1 or has a negative

Fiscal Externality. If $MVPF < 1$, the increase in government expenditure either has overall benefits to the beneficiaries of less than 1 or a positive Fiscal Externality.

As the EITC increases labor supply (one of the behavioral changes due to the policy), the fiscal externality of the EITC is negative (e.g. as take-up of other welfare programs is reduced), which results in a $MVPF > 1$. Other cash transfers in contrast often change behavior in the other direction (i.e., lead to lower labor supply) and thus have a negative fiscal externality, leading to a $MVPF < 1$.

One can further use the MVPF to directly compare two policies, at least if they have the same group of beneficiaries. Let's suppose we compare the EITC ($MVPF = 1.2/1.12$) with another cash transfer program that has a MVPF of 0.8. A budget neutral policy that took a dollar from the other cash transfer program and spent it on the EITC instead would be welfare enhancing. This is because one dollar spent on the EITC creates more utility than one dollar spent in the other cash transfer program, relative to the net cost to the government. This comparison only works if the group of beneficiaries for both policies are identical (i.e. working mothers). If the groups are not identical the social marginal utility of income of the policies' beneficiaries needs to be considered. A budget neutral policy change in favor of a policy with a lower MVPF could still enhance welfare if the average social marginal utility of income is high for this group of beneficiaries (e.g. the poor) and low for the other group of individuals benefiting from the policy change with the higher MVPF (e.g. the rich).

c) Briefly describe potential problems associated with the MVPF.

Solution: Potential problems may arise from the construction of the MPFV as one needs good estimates of WTP and government costs. Both the WTP and the cost of government may not be accessible. Hendren and Sprung-Keyser construct these values based on existing studies. Hence publication bias (only studies that find significant effects tend to be published and due to the way statistical tests are constructed, with 5% significance level, 5/100 studies on a topic will find significant effects although there are none) may cause problems for the use of the MVPF. Hendren and Sprung-Keyser try to account for this applying a method to test and correct the effect of publication bias on the observed set of estimates as a robustness check.

Additionally the authors note that many MVPF estimates for individual policies contain considerable sampling uncertainty. This is due to the sampling variation in the underlying estimates for the individual policies. As an example the 1990 health care expansion to children born after September 30, 1983 has a confidence interval ranging from 0.26 to infinity. Hence hypothesis testing leads to imprecise interpretations. To account for this the authors constructed pooled category averages of the policies under consideration.

Lastly the approach proposed by the authors does not judge whether outcomes are socially favorable. For instance, the MVPF for transfers to disabled children and their families is relatively low (0.76 for expanded eligibility for Supplemental Security Income at age 18). However, spending on these policies may increase social welfare, even though they have lower MVPFs as families with disabled children may have a high welfare weight. Decisions about optimal policy are determined by the welfare weights the government places on policy beneficiaries. If the government makes it a priority to provide support for disabled children, a transfer to those children could be welfare enhancing.

2) Read the Study "Value of a Statistical Life" by Knieser & Viscusi (2019) and answer the following questions. The study can be found here: [10.1093/acrefore/9780190625979.013.138](https://doi.org/10.1093/acrefore/9780190625979.013.138)

a) Briefly define the Value of a Statistical Life (VSL).

Solution: VSL describes estimates of willingness to pay for small reductions in mortality risks. These values are typically reported in units that match the aggregate dollar amount that a large group of people would be willing to pay for a reduction in their individual risks of dying in a year, so that one fewer death among the group during that year is expected on average.

b) Name and briefly explain the different approaches described in the study to measure a VSL.

Solution:

In researching the empirical content of the VSL the preferred choice of most researchers and U.S. policymakers is the revealed preference method wherein the researcher infers the willingness to pay for more or compensation required to accept less safety in the context of actual risk-taking decisions in labor and product markets.

- The universal estimating equation for VSL is a regression of wages that controls for other relevant personal characteristics that affect wages. The data can be derived from labor market surveys on workers' wages and risk of fatal injury. Usually regressions estimating the wage are semi-logarithmic. This means that the natural logarithm of wages is used as the independent variable. As depended variables typically demographic control variables (such as age and education) and other job characteristic variables (such as non-fatal injury risk, workers' compensation insurance coverage, and industry and occupation indicators) are used.
- The willingness to pay for safer products through higher prices for safety features, such as automobile airbags, is another way to estimate the VSL (revealed preferences). Corporate decisions concerning product safety and military strategy decisions concerning soldiers' combat deployments can also reveal aspects of the VSL. In such an example product market evidence is observed and used to estimate VSL. Among the complexities of VSL are the influence of psychological issues such as systematic biases in risk misperceptions, ambiguity aversion, rational ignorance of risk probabilities, loss aversion, and relative position effects stemming from where one is in the wage distribution.
- Most policy analyses in the United States favor primary reliance on revealed preference evidence in that the estimates reflect actual tradeoffs individuals make involving risks. Stated preference (sometimes referred to as contingent valuation) is a survey-based technique for establishing valuations. The subject is asked how much they value something. Stated preference studies have been used to provide estimates of the VSL both for traumatic fatality risks as well as for illnesses.

c) Name and discuss one problem of VSL described in the paper

Solution: One of the most important issues concerning the design of a study of the VSL relates to the role of the conceptual experiment. To calculate valid estimates of the VSL the change in fatality risk needs to be exogenous. One example of endogeneity is the choice of speed by a typical driver. Higher speed causes more casualties (higher risk) but the choice of speed is also influenced by the condition of the road (a good road constitutes lower risk). Therefore the relationship between speed and risk does not reveal anything about the risk trade-off posed by higher speeds.

Another is the issue of individuals' heterogeneity and the policy relevance accompanying it. Among the most prominent aspects of VSL heterogeneity are age variations in VSL and the income elasticity of VSL. The income elasticity is particularly important because it is the key to understanding international differences in VSL and the associated international differences in safety policy as well and indicating adjustments policymakers should make to threshold VSL values due to income growth.

Another major complication associated with estimating the VSL life is that the VSL can vary substantially with individual characteristics and risk types. Therefore, appropriate policy responses to threats to life and health may also need to vary with these factors. In reality, risks to life and health vary greatly with the type of risk (e.g. sudden fatal injury vs. prolonged illness) or the degree of latency (how far in the future the risk is).

Furthermore individuals have substantial biases and perceptual limitations when estimating the magnitudes of small probabilities. Because the VSL is calculated by observing what people are willing to give up to reduce risks, if individuals misperceive risks, VSL estimates based on those risk perceptions will be either imprecisely estimated or biased.

2 Application of General Equilibrium

3) Individuals A and B of a two good exchange economy (x_1, x_2) have the following utility functions:

$$U_a(x_{a1}, x_{a2}) = \ln x_{a1} + 2 \ln x_{a2}$$

$$U_b(x_{b1}, x_{b2}) = 2 \ln x_{b1} + \ln x_{b2}$$

Let the initial endowments of the individuals be $R_A = (9, 3)$ and $R_B = (12, 6)$.

- a) Find the excess demand for each good, give a definition of Walras' law and verify that the sum of the excess demands adds up to 0.

Solution:

To calculate the excess demand we have to solve the maximization problem. The Lagrange function for Individual A looks as follows:

$$\mathcal{L}(x_1^a, x_2^a, \lambda^a) = \ln(x_1^a) + 2 \ln(x_2^a) + \lambda^a (9p_1 + 3p_2 - p_1 x_1^a - p_2 x_2^a) \quad (1)$$

The first order conditions are:

$$\frac{\partial \mathcal{L}}{\partial x_1^a} = \frac{1}{x_1^{*a}} - \lambda^{*a} p_1 = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial x_2^a} = \frac{2}{x_2^{*a}} - \lambda^{*a} p_2 = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 9p_1 + 3p_2 - p_1 x_1^{*a} - p_2 x_2^{*a} = 0 \quad (4)$$

After setting the equations (2) and (3) equal and substituting the values for x_1 and x_2 in equation (4) for individual A the demand functions are:

$$x_1^{*a} = 3 + \frac{1}{p}; \quad x_2^{*a} = 6p + 2$$

For Individual B the demand functions are:

$$x_1^{*b} = 8 + \frac{4}{p}; \quad x_2^{*b} = 4p + 2$$

By using the previous results the surplus demands can be calculated:

$$E_1 = \frac{5}{p} - 10$$

$$E_2 = 10p - 5$$

Verifying that Walras' law holds, the weighted sum of excess demand must be zero:

$$\begin{aligned}pE_1 + E_2 &= 0 \\p\left(\frac{5}{p} - 10\right) + 10p - 5 &= 0 \\5 - 10p + 10p - 5 &= 0\end{aligned}$$

It can be seen that Walras' law holds. In equilibrium, the excess demand for each good must be zero.

b) Calculate the equilibrium prices of the goods.

Solution:

The equilibrium price ratio is determined by solving $E_1 = 0$:

$$\frac{5}{p} - 10 = 0 \Leftrightarrow p = \frac{1}{2} \quad (5)$$

c) Determine the allocation that results in equilibrium.

Solution:

The price ratio p is inserted into the demands to determine the equilibrium allocation:

$$\begin{aligned}x_1^{*a} &= 5, \quad x_2^{*a} = 5 \\x_1^{*b} &= 16, \quad x_2^{*b} = 4\end{aligned}$$